

More on conditional expectations

In this assignment, we explore conditional expectations and their calculations for the random walk stochastic process. We will gradually develop all the ingredients needed to compute conditional expectations given the information contained in some σ -algebra. A general sample space will be denoted Ω , with more specific notation like Ω_N reserved for sample spaces involving coin tosses.

The probability space for the random walk stochastic process is the triple $(\Omega_N, \mathcal{F}, \mathbb{P})$ where Ω_N is the sample space of sequences $\underline{\omega} = \omega_1 \cdots \omega_N$ of N coin tosses, $\mathcal{F}$ is the σ -algebra corresponding to the power set of Ω_N , and the probability measure $\mathbb{P}$ corresponds to assigning a probability of p for heads (H) and q for tails (T) with $p + q = 1$.

The stochastic process $\mathbb{M}_n$, $0 \leq n \leq N$, is defined as follows. Define $\mathbb{M}_0 = 0$. Then, we define the independent random variables $\mathbb{X}_i$ which depend only on the i^{th} coin toss, specified by

$$\mathbb{X}_i(\omega_i) = \begin{cases} +1 & \omega_i = H, \\ -1 & \omega_i = T. \end{cases} \quad (1)$$

The random variable $\mathbb{M}_n$ is then defined as the sum of the first n $\mathbb{X}_i$:

$$\mathbb{M}_n(\omega_1 \cdots \omega_n) = \sum_{i=1}^n \mathbb{X}_i(\omega_i), \quad \text{for } n = 1, \dots, N. \quad (2)$$

We also need the definition of an event. An event is a subset of the sample space defined by a rule or a collection of rules that the elements of the subset must satisfy. For instance, the event $\mathbb{M}_1 = 1$ is a subset of Ω_N containing sequences of coin tosses which start with a H .

Note: Exercises 1a, 1d, 1e assume that the total number of coin tosses is $N = 4$. Exercises 1b, 1f, 1g, 1h assume that $N = 3$. Exercises 1c, 1i, 1j have to be worked out for a general N .

Exercise 1a (4 points): Describe the σ -algebras $\sigma(\mathbb{M}_2)$ and $\sigma(\mathbb{M}_3)$ generated by the random variables $\mathbb{M}_2$ and $\mathbb{M}_3$ respectively.

Recall that the σ -algebra generated by a random variable $\mathbb{Y}$ contains subsets of the sample space of the form $\{\underline{\omega} \in \Omega \mid \mathbb{Y}(\underline{\omega}) \in I\}$ where I runs through all Borel subsets of $\mathbb{R}$. A shorthand notation for such a set is $\{\mathbb{Y} \in I\}$.

When we have discrete random variables, this definition can be simplified. Consider the subsets (equivalently, events) of the form $\{\underline{\omega} \in \Omega \mid \mathbb{Y}(\underline{\omega}) = a\}$, alternatively called $\{\mathbb{Y} = a\}$, where a is one of the finite number of values $\mathbb{Y}$ is allowed to take. To form the σ -algebra $\sigma(\mathbb{Y})$, take the collection of events $\{\mathbb{Y} = a\}$ obtained by running through all possible values

of a and then take complements, countable unions and intersections. For instance, when the total number of coin tosses is $N = 4$, the event $\{\mathbb{M}_2 = 2\}$ consists of the sequences $\{HHHH, HHHT, HHTH, HHTT\}$, i.e., all sequences starting with two heads.

For the purposes of this question, you can take the total number of coin tosses as $N = 4$. It is enough to give the basic sets of the type $\{\mathbb{Y} = a\}$ for all allowed values of a and not all sets in the σ -algebra. ■

Exercise 1b (6 points): Let $\mathcal{A}_n$ be the σ -algebra generated by the random variables $\mathbb{M}_1, \mathbb{M}_2, \dots, \mathbb{M}_n$. That is, $\mathcal{A}_n$ is obtained by taking complements, countable unions and intersections of all events of the form $\{\mathbb{M}_i = a\}$ for all $1 \leq i \leq n$ and all allowed values of a . Compute $\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3$. For the purposes of this question, you can take the total number of coin tosses to be $N = 3$.

Let us define $\mathcal{A}_0$ to be the trivial σ -algebra $\{\emptyset, \Omega\}$. Show that the sequence of σ -algebras $\{\mathcal{A}_n\}_{0 \leq n \leq 3}$ is a filtration. That is, show that $\mathcal{A}_0 \subseteq \mathcal{A}_1 \subseteq \mathcal{A}_2 \subseteq \mathcal{A}_3$ is a chain of σ -subalgebras. ■

A filtration like $\{\mathcal{A}_n\}_{0 \leq n \leq N}$ is called a *natural filtration* associated to the random process $\mathbb{M}_n$ since it is directly defined in terms of the events associated to the random process. We will use this natural filtration later in the assignment.

Note: We frequently drop the extra qualification that $0 \leq n \leq N$ in the subscript of the filtration and simply write $\{\mathcal{A}_n\}$. An $\mathcal{A}_n$ without the enclosing braces will be used to refer the n^{th} σ -algebra in the filtration.

Next, we look at expectation values. Start with our usual definition of expectation values of a random variable:

$$\mathbb{E}[\mathbb{Y}] = \sum_{\omega \in \Omega_N} \mathbb{Y}(\omega) \mathbb{P}\{\omega\} . \quad (3)$$

One can rearrange the terms in the above sum to convert it to a sum over the values of the random variable $\mathbb{Y}$. Let the set of values that the random variable $\mathbb{Y}$ be $\text{Ran}(\mathbb{Y})$, the *range* of $\mathbb{Y}$. Then, we can write

$$\mathbb{E}[\mathbb{Y}] = \sum_{y \in \text{Ran}(\mathbb{Y})} y \mathbb{P}\{\mathbb{Y} = y\} . \quad (4)$$

That is, we sum over the values of the random variable $\mathbb{Y}$ weighted by its probability distribution $\mathbb{P}\{\mathbb{Y} = y\}$.

Exercise 1c (4 points): Demonstrate this result for the random variable $\mathbb{M}_n$. That is, show that

$$\sum_{\omega \in \Omega_N} \mathbb{M}_n(\omega) \mathbb{P}\{\omega\} = \sum_{k=-n}^n k \mathbb{P}\{\mathbb{M}_n = k\} . \quad (5)$$

Hint: The individual probability for a sequence of n coin tosses with n_H heads and n_T tails is $p^{n_H}q^{n_T}$ with $n = n_H + n_T$ and $k = n_H - n_T$. Club together all terms in the sum over sequences that give the same value of k first, and then sum over all such k . As usual, you must handle the coin tosses from $n + 1$ to N appropriately.

By the same kind of argument, show that the above is true for any function $f(\mathbb{M}_n)$ of the random variable:

$$\mathbb{E}[f(\mathbb{M}_n)] = \sum_{\underline{\omega} \in \Omega_N} f(\mathbb{M}_n)(\underline{\omega}) \mathbb{P}\{\underline{\omega}\} = \sum_{k=-n}^n f(k) \mathbb{P}\{\mathbb{M}_n = k\} . \quad (6)$$

■

Let us now look at conditional expectations. These are defined based on conditional probabilities. Suppose we have events A and B and we want to compute the probability that A occurs given that B has occurred. This means we have to restrict our attention to that part of the sample space in which B has occurred. Recall that an event such as A or B correspond to certain subsets of the sample space. For instance, if we say B is the event that the first two coin tosses are H and T respectively, we have to restrict ourselves to the subset $B = \{\underline{\omega} \in \Omega_N \mid \omega_1 = H, \omega_2 = T\}$. Similarly, an example of A would be the event that the first three coin tosses are HTH .

Once we restrict ourselves to the subset B , we then look for occurrences of the event A within the subset B rather than the entire sample space. Clearly, this requires counting the number N_{AB} of joint occurrences of the event A and B , and the probability of occurrence of A given B is this number N_{AB} divided by the number N_B of occurrences of the event B . That is,

$$\mathbb{P}\{A \mid B\} = \frac{N_{AB}}{N_B} = \frac{N_{AB}/N_{\text{tot}}}{N_B/N_{\text{tot}}} = \frac{\mathbb{P}\{AB\}}{\mathbb{P}\{B\}} . \quad (7)$$

In the last step, N_{tot} is the total number of events in the sample space (for the coin-toss sample space, we have $N_{\text{tot}} = 2^N$), and we have rewritten the right hand side in terms of the probabilities of the joint event AB and the event B . When A is the event that the first three coin tosses are HTH and B is the event that the first two coin tosses are HT , we have

$$N_B = 2^{N-2} , \quad N_{AB} = 2^{N-3} , \quad \text{giving} \quad \mathbb{P}\{A \mid B\} = \frac{2^{N-3}}{2^{N-2}} = \frac{1}{2} . \quad (8)$$

On the contrary, if A is the event that the first three coin tosses are TTH and B is the event that the first two coin tosses are HT , these events cannot jointly occur. So, N_{AB} in this case is zero and hence $\mathbb{P}\{A \mid B\} = 0$.

In other words, one has to recompute the probability of occurrence of an event A given new information that the event B has occurred. This new probability is the conditional

probability $\mathbb{P}\{A \mid B\}$ and is typically different from the original probability of occurrence of A which is simply $\mathbb{P}\{A\}$.

Another example which will be useful in defining conditional expectations is when we take the event A to be a singleton set, i.e., a set that contains a single element of the sample space. The original probability of any sequence in the sample space is $1/2^N$ for the symmetric walk. Let us again take B to be the subset of sequences starting with HT . If the sequence $\underline{\omega}$ (whose probability we are recomputing) is of the form $HT\omega_3 \cdots \omega_N$, then the joint probability of the sequence and B occurring is the original probability itself, so that

$$\mathbb{P}\{\underline{\omega} \mid B\} = \frac{\mathbb{P}\{\underline{\omega}\}}{\mathbb{P}\{B\}} = \frac{1/2^N}{1/2^2} = \frac{1}{2^{N-2}} . \quad (9)$$

If the sequence $\underline{\omega}$ does not start with HT , then it does not belong to B , and in particular the joint probability of $\underline{\omega}$ and B occurring is zero. In that case, the recomputed probability is zero. Thus, we have the revised probabilities

$$\mathbb{P}\{\underline{\omega} \mid B\} = \begin{cases} \frac{1}{2^{N-2}} & \underline{\omega} \in B , \\ 0 & \underline{\omega} \notin B . \end{cases} \quad (10)$$

Again, we see that the probability of occurrence of a sequence starting with HT is enhanced to $1/2^{N-2}$ from the original $1/2^N$, whereas the probability of occurrence of a sequence not of this form has diminished to zero.

The expectation value of a random variable $\mathbb{Y}$ given the knowledge that an event $B \subseteq \Omega_N$ has occurred is defined as

$$\mathbb{E}[\mathbb{Y} \mid B] = \sum_{\underline{\omega} \in B} \mathbb{Y}(\underline{\omega}) \mathbb{P}\{\underline{\omega} \mid B\} . \quad (11)$$

There are two changes compared to the formula for a full expectation value: (1) the range of elements of the sample space is restricted to the subset B , (2) the probability associated to each such element $\underline{\omega}$ is $\mathbb{P}\{\underline{\omega} \mid B\}$ and is typically different from the original probability $\mathbb{P}\{\underline{\omega}\}$.

Now, we recast the formula (11) for the conditional expectation value above in terms of the values of the random variable Y . One must now sum over only those values of $\mathbb{Y}$ that arise from evaluating $\mathbb{Y}$ on elements of the event B rather than the full sample space. The set of such values obtained by evaluating $\mathbb{Y}$ on the elements in the event B is written as $\text{Ran}_B(\mathbb{Y})$,

$$\mathbb{E}[\mathbb{Y} \mid B] = \sum_{y \in \text{Ran}_B(\mathbb{Y})} y \mathbb{P}\{\mathbb{Y} = y \mid B\} . \quad (12)$$

It is clear from the above formula that the conditional expectation value is fixed once the event B is specified. If the event B is changed to another event, then the expectation value may also change.

Thus, the conditional expectation value $\mathbb{E}[\mathbb{Y} \mid B]$ is a function of the event B .

Exercise 1d (4 points): Let us go back to the random walk. Let the event B be the subset of the sample space Ω_N consisting of sequences $\underline{\omega}$ with $\mathbb{M}_3 = 1$. That is,

$$B = \{\underline{\omega} \in \Omega_N \mid \mathbb{M}_3(\underline{\omega}) = 1\} , \quad (13)$$

also written in short as $B = \{\mathbb{M}_3 = 1\}$. Compute the conditional expectation value

$$\mathbb{E}[\mathbb{M}_4 \mid B] \quad \left(\text{also written as } \mathbb{E}[\mathbb{M}_4 \mid \mathbb{M}_3 = 1] \right) , \quad (14)$$

by first using the formula (11), and then separately using (12), and check that you get the same values. You should compute all the required probabilities that appear in the formulas (11) and (12). For the purposes of this question, you can take the number of tosses to be $N = 4$ and borrow any required results from previous exercises. ■

As discussed above, the conditional expectation $\mathbb{E}[\mathbb{Y} \mid B]$ is a function of the event B . We now progress one stage further in abstraction and consider the smallest σ -algebra that contains the event B , usually denoted as $\sigma(B)$. This is clearly the σ -algebra which contains B , its complement B^c , the null set $\emptyset$ and the full sample space Ω and nothing else. That is,

$$\text{Smallest } \sigma\text{-algebra containing } B = \sigma(B) = \{B, B^c, \emptyset, \Omega\} . \quad (15)$$

It is easy to check that the above collection of subsets is closed under unions, intersections and complements. The reason to consider this extra layer of abstraction is that, if one knows that the event B occurs, one also immediately knows that the event B^c does **not** occur. Further, we know trivially that $B \cap B^c = \emptyset$ is the null event and $B \cup B^c = \Omega$ is the event that contains all possibilities. Together, this σ -algebra completely summarizes the new information that *can be* specified once the event B is specified – either we say B occurs, or B^c occurs, or $B \cap B^c = \emptyset$ occurs, which is the same as the information that no sequence in the sample space has occurred, or finally $B \cup B^c = \Omega$ occurs, which is the same as having no prior information about any specific event occurring.

Exercise 1e (6 points): One can now compute the conditional expectation value of a random variable $\mathbb{Y}$ given each of the subsets in the σ -algebra $\sigma(B)$. Take $\mathbb{Y} = \mathbb{M}_4$, $B = \{\mathbb{M}_3 = 1\}$ as in the previous exercise and compute

$$\mathbb{E}[\mathbb{M}_4 \mid B^c] , \quad \mathbb{E}[\mathbb{M}_4 \mid \emptyset] , \quad \mathbb{E}[\mathbb{M}_4 \mid \Omega] . \quad (16)$$

For the purposes of this question, you can take $N = 4$. You can use whichever of the two formulas (11), (12) you find convenient to compute the above conditional expectation values. ■

Thus, we have four values for the conditional expectation and clearly it is a function of the events present in the σ -algebra $\sigma(B)$.

The function on the events present in $\sigma(B)$ obtained above is denoted as $\mathbb{E}[\mathbb{Y} \mid \sigma(B)]$ and its values are given by the condition expectation values $\mathbb{E}[\mathbb{Y} \mid A]$ with $A \in \sigma(B)$.

Note on notation: Unfortunately, the notation $\mathbb{E}[\cdot \mid \cdot]$ is used for many different but closely related quantities. The first argument is usually a random variable in all these different uses. The second argument is either an event (which is a single subset of the sample space), or a σ -algebra of subsets of the sample space. When the second argument is an event B , then $\mathbb{E}[\mathbb{Y} \mid B]$ is a *number*. However, when the second argument is a σ -algebra $\mathcal{G}$, the quantity $\mathbb{E}[\mathbb{Y} \mid \mathcal{G}]$ is a *function* on the events in the σ -algebra $\mathcal{G}$.

Interpreting the conditional expectation $\mathbb{E}[\mathbb{Y} \mid \sigma(B)]$ as a random variable on the sample space: As we have discussed above, the conditional expectation $\mathbb{E}[\mathbb{Y} \mid \mathcal{G}]$ is a function on the events that are present in the σ -algebra $\mathcal{G}$. This can be easily also considered as a function on the sample space since every element of the sample space is an element of one of the events in the σ -algebra $\mathcal{G}$. More precisely, since we are allowed to take arbitrary countable unions and intersections inside a σ -algebra, one can come up with a partition of the sample space into subsets $B_1, B_2, \dots, B_p$ where the subsets are all non-overlapping ($B_i \cap B_j = \emptyset$) and their union gives the sample space ($\bigcup_{i=1}^p B_i = \Omega$). Once we have such a partition, we give the value of $\mathbb{E}[\mathbb{Y} \mid \mathcal{G}]$ on an element $\underline{\omega}$ of the sample space by (1) finding the unique event B_i in the partition that contains this element $\underline{\omega}$, and (2) assigning the value that $\mathbb{E}[\mathbb{Y} \mid \mathcal{G}]$ has on this event. Thus, $\mathbb{E}[\mathbb{Y} \mid \mathcal{G}]$ is a random variable that is defined on the same sample space as the random variable $\mathbb{Y}$.

For instance, suppose B is an event, e.g., the event consisting of all coin toss sequences starting with HT . The σ -algebra generated by B is $\{B, B^c, \emptyset, \Omega\}$. Clearly, the sets B and B^c partition the sample space since $B \cap B^c = \emptyset$ and $B \cup B^c = \Omega$. Thus, the value of the random variable $\mathbb{E}[\mathbb{Y} \mid \sigma(B)]$ on the elements $\underline{\omega} \in \Omega$ are given by

$$\mathbb{E}[\mathbb{Y} \mid \sigma(B)](\underline{\omega}) = \begin{cases} \mathbb{E}[\mathbb{Y} \mid B] & \text{if } \underline{\omega} \in B, \\ \mathbb{E}[\mathbb{Y} \mid B^c] & \text{if } \underline{\omega} \in B^c. \end{cases} \quad (17)$$

Exercise 1f (4 points): Compute the random variable $\mathbb{E}[\mathbb{M}_3 \mid \sigma(\mathbb{M}_2)]$. For the purposes of this exercise, you can take the sample space to consist of sequences of three coin-tosses only. The events in the σ -algebra $\sigma(\mathbb{M}_2)$ are simpler to describe in the case $N = 3$ compared to the $N = 4$ result in Exercise 1a.

We have discussed in class that this random variable is $\mathbb{M}_2$ itself, but I want you to reach this conclusion systematically. This means, first write down the events A in the σ -algebra $\sigma(\mathbb{M}_2)$. Next, compute the conditional expectation values $\mathbb{E}[\mathbb{M}_3 \mid A]$ for each event A in $\sigma(\mathbb{M}_2)$ so that we get a function on the events in the σ -algebra. Finally, come up with a partition of the sample space using the events in $\sigma(\mathbb{M}_2)$, and reinterpret $\mathbb{E}[\mathbb{M}_3 \mid \sigma(\mathbb{M}_2)]$ as a function on the sample space by using this partition of sample space. ■

Exercise 1g (4 points): Compute the random variable $\mathbb{E}[\mathbb{M}_3 \mid \mathcal{A}_2]$ where $\mathcal{A}_2$ is the σ -algebra generated by the random variables $\mathbb{M}_1, \mathbb{M}_2$ (see Exercise 1b). How is this answer different from the answer of Exercise 1f? ■

Exercise 1h (4 points): Let $f(x)$ be an ordinary function of one variable. Then $f(\mathbb{M}_3)$ is a random variable since it is a function of the random variable $\mathbb{M}_3$. Compute the conditional expectation $\mathbb{E}[f(\mathbb{M}_3) \mid \sigma(\mathbb{M}_2)]$ and show that it is a random variable which is a function $g(\mathbb{M}_2)$ where g is an ordinary function that is expressed in terms of f .

Based on your calculations in Exercises 1f, 1g and the current exercise, and without further computation, comment on whether the conditional expectations $\mathbb{E}[f(\mathbb{M}_3) \mid \sigma(\mathbb{M}_2)]$ and $\mathbb{E}[f(\mathbb{M}_3) \mid \mathcal{A}_2]$ are the same or different. ■

Exercise 1i (2 points): In fact, given a time instant n with $n < N$, we can show that $\mathbb{E}[f(\mathbb{M}_N) \mid \mathcal{A}_n] = f_n(\mathbb{M}_n)$ where f is some function of one variable of our choice and f_n is a function which can be obtained in terms of f by computing the conditional expectation above. In words, argue why you expect the above general result to be true. The argument is a simple extension of the calculations above with $N = 3$. ■

Suppose $\mathcal{G}$ and $\mathcal{H}$ are σ -algebras such that $\mathcal{H} \subseteq \mathcal{G}$, that is, $\mathcal{H}$ is a σ -subalgebra of $\mathcal{G}$. Then, we have the following result called the property of iterated conditioning:

$$\mathbb{E}\left[\mathbb{E}[\mathbb{Y} \mid \mathcal{G}] \mid \mathcal{H}\right] = \mathbb{E}[\mathbb{Y} \mid \mathcal{H}] . \quad (18)$$

Exercise 1j (2 points): Using the property of iterated conditioning by choosing $\mathcal{G}$ and $\mathcal{H}$ appropriately from the filtration $\{\mathcal{A}_n\}$, show that the stochastic process $f_n(\mathbb{M}_n)$, $0 \leq n \leq N$, is a martingale with respect to the filtration $\{\mathcal{A}_n\}$. ■